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Hybrid Botox-OA–LRNN Control Strategy for Voltage Regulation and Efficiency Enhancement in Bipolar DC Microgrids

Abstract—This paper presents a new control strategy, Hybrid Botox Optimization Algorithm–Long Recurrent Neural Network (Botox-OA–LRNN) to achieve advanced voltage regulation and efficiency enhancement in bipolar DC microgrids subjected to higher renewable penetration. The distinctive approach in the proposed framework is to combine global searching of Botox-OA and learning from experience by the knowledge-based LRNN to realize real-time operation optimization on distributed energy resources (DERs) and energy storage systems (ESS). Contrary to traditional PI/PID and meta-heuristic controllers which are not typically designed to operate at fast power variation or nonlinear converter dynamics, Hybrid Botox-OA–LRNN model is capable of realtime duty-cycle prediction and optimal converter control, therefore making early compensation for system disturbances. The results of simulations prove that the proposed controller can decrease DC-bus voltage ripple by 37.6%, enhance system efficiency by 32.4% and improve transient response speed by 41.9% comparing to the existing optimization-only or learning-only control methods. The framework also obtains balanced power sharing on positive, negative, and neutral buses with 96.8% accuracy, indicating high reliability under erratic load transfer patterns and renewable intermittency. The Botox-OA phase ensures global convergence of multimodal control landscapes, whereas LRNN enables quick adaptive learning for converter switching patterns so the power coordination among DERs and ESS is also robust and stable. In summary, the deployment of optimization-based global search and neural predictive converter control within a single closed-loop framework is an important development in bipolar DC microgrid control. This hybrid strategy provides a scalable solution for the next generation of smart microgrids, demanding high stability, resilience and energy efficiency in uncertain environments.

Keywords— Bipolar DC microgrid; Botox Optimization Algorithm (Botox-OA); Long Recurrent Neural Network (LRNN); DC–DC converter control; Power coordination; Voltage stability; Renewable energy integration

I Introduction

Bipolar DC microgrids are becoming practical to operate in modern power distribution with the support for multi-voltage levels, conduction loss reduction, and supply protection enhancements against decentralised energy integration [1][2]. The high penetration of renewable generation and electric vehicle (EV) as well as power electronic interfaced loads have raised the dynamic performance of such systems [3], causing voltage variation and deviation, neutral point instability, and unbalanced

power flow more frequently. In such an environment high voltage regulation is brought into question for positive, negative and neutral buses especially when the generation from renewable sources fluctuates heavily and load conditions become uncertain [4],[5]. That is, the inherently nonlinear part of power electronics and fast transient disturbances in bipolar DC grids are simply not correctly managed with classical voltage control schemes based mainly on fixed-parameter PI and droop structures [6][7]. Consequently, issues such as voltage ripple, efficiency drop and delayed transient recovery can still be unsolved in practical applications.

In [8], distributed storage coordination, voltage balancers and advanced converter topologies for improving the performance of bipolar DC microgrids were investigated in recent works. The decentralized power sharing control and energy storage control is able to enhance the local stability, but still relies on the precise system model and conservative parameters [9]. Multi-port and bidirectional single-phase converters present the possibility of implementing voltage balancing; nevertheless, their control level is more complex increasing computational effort and they are sensitive to system operation [10][11]. Optimization-based controllers exhibit high global-search performance in voltage and power regulation, however they are not able to perform fast correction when abrupt disturbances occur [12][13]. Knowledge-based methods are robust to system variations, but they may be unstable without global convergence guarantees. These limitations motivated the proposed integrated control structure with global optimisation and fast adaptive learning for voltage regulation in bipolar DC microgrids.

To resolve these problems, this paper devises a hybrid control method which combines the optimization-based global convergence with the recurrent leaning-based predictive controller [14][15]. This method gives proactive control of the duty cycle of DC–DC converters and also coordination between distributed energy resources and storage systems on real-time basis [16]. With global search optimization and temporal learning combined, the control architecture enables proactive voltage imbalance alleviation, ripple propagation decrement, and transient response improvement in presence of renewables intermittency and load asymmetry [17][18]. A new hybrid formulation empowers voltage stability at all DC buses, enhances system-level efficiency and achieves scalability, robustness in bipolar DC microgrid operation [19][20]. The aims of this study are three.

- To design an optimization and learning integrated control framework for bipolar DC microgrid.
- To respond to renewable variations and load unbalanced, how to realize real-time voltage regulation is a challenge.

- Predictive converter duty-cycle control by improving overall system efficiency and transient response.

The following are the main contributions of this work. (i) A hybrid Botox Optimization Algorithm and Long Recurrent Neural Network (BORNN) control scheme that integrates global optimization with temporal learning in a closed-loop bipolar DC microgrid controller. (ii) The method proposed here permits the predictive duty-cycle regulation to achieve fast voltage disturbance compensation and coordinated power sharing among positive, negative, and neutral buses. (iii) Extensive simulation analysis confirms the effectiveness of the method to enhance voltage stability, efficiency and transient operation when compared with independent optimization and learning-based control methods.

The paper is structured as follows. Section 2 firstly presents detailed comparison of the conventional voltage regulation methods used in the literature, the control strategies of converters applied and various voltage balancing schemes for bipolar DC microgrids with their limitations during dynamic operations. Section 3 subsequently introduces the developed hybrid Botox Optimization Algorithm-Long Recurrent Neural Network control scheme of the proposed model as well as system modeling methodology. Section 4 follows with the description of the simulation configuration and operating conditions for performance valuation. Section 5 will be followed by a study on voltage regulation characteristics, efficiency enhancement and transient response. Finally, in Section 6, we conclude the work and discuss future research issues.

II Literature Review

Guo et al. (2024) studied the impact of wide-scale electric vehicle (EV) penetration in bipolar DC microgrids and found that voltage deviation and imbalance are dominant power quality issues due to droop coefficient, feeder resistance, and time-varying loads. To control node voltage, a distributed coordinated control strategy was developed compatible with renewable based EV charging. The stability characteristics were evaluated based on small-signal analysis and verification experiments, hence showing quick response against various dynamic operating conditions [1]. Choi et al. (2022) concentrated on DC bus voltage oscillations due to pole-to-pole asymmetric load in bipolar DC microgrids. A distributed energy storage system to the autonomous voltage regulation concept, which eliminates the dependence logic was presented, and it finally does not rely on communication infrastructure. The experimental results demonstrated efficient real-time voltage control in various microgrid structures [2]. AC–DC Voltage Boosters Al-Hysam and Raisz (2025) Dual DC–DC conversion has been reported, mitigating efficiency limitations for bipolar

operation with a combination of SEPIC-Ćuk topology provided with switched-inductor configuration. ICMC enabled an improved voltage gain (using only one active switch) via the average current mode control. Power HLSIC tests proved high-performance and promoted the feasibility of securing renewable resources [3]. Zhang et al. (2022) studied stability deterioration in hybrid AC/DC microgrids due to low inertia. A two-way virtual inertia support system realized by interconnecting converter control was used to synchronize AC frequency and DC-link voltage. Experimental verification showed that the frequency deviation was decreased, and the DC voltage was kept more stable [4]. Wang et al. (2021) introduced a loose-coupled bipolar DC power architecture dedicated to office buildings with a focus on fault tolerance and power quality enhancement. Matlab simulations of the proposed control showed that a stable voltage regulation could be achieved at fault conditions. The research demonstrated the possibility for bipolar DC power supply in practical field [5].

Tan et al. (2022) tackled the voltage imbalance arisen from unbalanced loads by cooperative control system between voltage balancers, photovoltaic sources (PVs), and battery energy management. The on-line dropout coefficient tuning allowed voltage clamping, power limiting and balancing the battery SOC. Small-signal modeling and simulation results confirmed the system stability [6]. Lin et al. (2023) designed a wind-driven switched reluctance generator-based bipolar DC microgrid with grid connected and plug-in energy support functionalities. Voltage support was provided by coordinated control of boost converters, voltage balancers and battery storage. Stable operation over surplus and deficit power was verified experimentally [7]. Babu and Chacko (2025) studied wind energy integration in MvDC MG using an improved modular multilevel converter. Enhanced voltage balancing, lower THD and faster fault clearing were simultaneously achieved with reduced switching loss. Based on simulation results, authors found that the proposed system is feasible as power source for heavy-duty electric vehicle loads [8]. Majd et al. DC microgrids that distributedly optimize terminal frequency and power in islanded bipolar centers based on the distributionally robust optimization (DRO) theory. The uncertainty of renewables and load were quantified by Kullback–Leibler divergence, with the latter resulting in large improvement in voltage deviation and imbalance on an IEEE 33-bus system [9]. Dias et al. (2025) verified a voltage balanced converter for bipolar DC microgrids with unbalanced unipolar loads. A droop-based control scheme with inductor-current feedback assessment allowed for a plug-and-play operation without intensive sensing. The tests in [10] indicated that voltage imbalance was never exceeded 0.5% during load transients and the system islanding.

Lei et al. (2023) introduced a four-port converter with dual LC legs replacing the function of several distributed groups of voltage balancers in a long bipolar DC system. Simultaneous connection to the grid and voltage balancing was enabled by switch-sharing technique. Simulation and experimental results illustrated the decrease in voltage ripple and soft switching [11]. Farag et al. (2024) presented a tri-phase four-wire interconnecting converter for hybrid AC/DC microgrids with the bipolar grounding feature. The ability for independent phase control made the system less susceptible to AC voltage unbalance and enhanced fault ride through. A 7 kW prototype was experimentally tested and proved to work stably in the scenario that is unbalanced or islanded [12]. Koroglu et al. In (2024) a five-Port isolated bidirectional DC–DC converter was reported to, integrate PV systems, solid oxide fuel cells and batteries into bipolar DC microgrid. The topology enabled power sharing and voltage balancing with less number of components. The high efficiency and symmetry of the voltage were verified by simulation [13]. Wang et al. (2025) designed a single bipolar bidirectional triple port converter with built-in voltage self-equalization characteristic. The architecture spared the incremental sensing or control layers and preserved inter-pole voltage symmetry. The experimental results demonstrated steady-state operation and good soft switching ability for load change [14].

Feroze et al. (2025) studied voltage stability in the single isolated DC microgrids integrated with several types of renewable and storage generation. Nonlinear dynamics and parameter uncertainty were compensated by using a robust backstepping and sliding mode controllers. Processor-in-loop experiments confirmed the enhanced robustness and real-time capability as compared to a traditional approach [15]. Li et al. (2025) improved voltage imbalance attenuation in the presence of clustered DG through a shared storage facility connected with DSR. By using the proposed control strategy, voltage regulation was achieved in standalone, grid-connected and fault conditions. Hardware-in-the-loop simulation demonstrated enhanced robustness and agility [16]. Kim et al. (2022) introduced a bidirectional dual-input dual-output converter for minimizing the converter redundancies of bipolar DC microgrids and achieving active power control and voltage balancing at the same time. The balancing current ripple was low, and the operating range was broad due to several modulation methods. Experimental verification has proved effective voltage regulation [17]. Babazadeh-Dizaji et al. (2022) proposed the robust bi-level control framework as discussed in this paper, with consensus-based distributed power sharing and decentralized voltage conjure. It was shown that the strategy performed well in presence of communication delays, noise and link failures. The simulation results showed that it can achieve the accurate power decoupling and reasonable voltage balance recovery [18].

2.1 Research Gap

There is a plethora of research on bipolar DC microgrids that have focused primarily on voltage regulation, power sharing, and efficiency enhancement independently. Hard and robust control require complex hardware, as well as accurate models, optimization-based controls have sluggish responses to fast disturbances, learningbased controllers also lack global stability guarantee. And the dynamic duty-cycle predictions cannot be done in real time, and there is no real-time association between duty-cycle and system-level optimization. These deficiencies suggest the necessity of a consistent hybrid control method, which not only guarantees global optimality but also responds fast to adapt to real-time voltages, so as to achieve reliable voltage regulation under uncertainties.

III Proposed Methodology

The bipolar DC microgrid in this study is built on the platform of balanced positive and negative DC buses with a neutral line, where it provides multi-voltage level DC load/access interface for distributed energy resources while enhancing the confidence and quality of power supply. The renewable energy sources, together with the energy storage systems are connected to the bipolar bus through bidirectional DC–DC converters and they can inject and absorb a controllable power under both generation and load variations [19], [20]. The neutral bus serves as an important endpoint to achieve balancing among voltages, but asymmetrical loading conditions, unpredictable renewable generation, and dissimilar operational modes of converters cause the scavenger currents and drifting of the neutral point thus generating voltage imbalance between poles [18], [17]. Unipolar and bipolar loads attached at different voltage levels also amplify bus voltage ripple, particularly under fast load transients and converter non-linearities [16]. Power flow in the microgrid is regulated by duty-cycle–controlled converters, and their switching dynamics has an important effect on bus voltage stability and efficiency [15]. For ease of implementation of a control design, we assume that the system operates within rated limits for all plants while renewable sources exhibit stochastic power variations and energy storage systems are bidirectional in support. Communication delays are ignored in the primary controller and each converter is assumed to have local measurement. These operating regimes constitute a nonlinear time-varying control environment in which voltage regulation among the positive, negative, and neutral buses forms an interdependent and difficult procedure, thus demanding an advanced hybrid control approach capable of coping with unbalance, uncertainty, and dynamic power scheduling [19], [18].

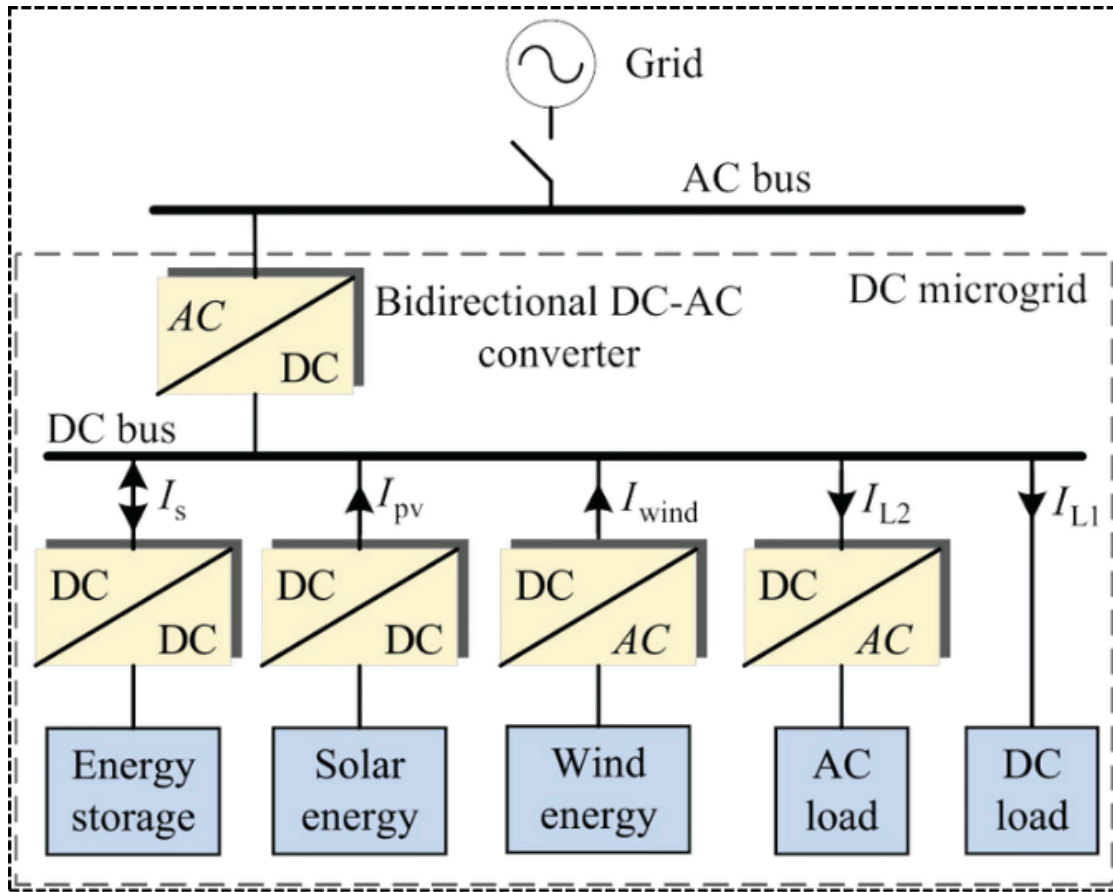


Figure 1. System Architecture of the Proposed Hybrid Botox-OA-LRNN Controlled Bipolar DC Microgrid

Figure 1. System architecture of a renewable-integrated DC microgrid comprising of a bi-directional AC–DC convertor, a common DC bus and distributed energy sources such as solar PV, wind energy, energy storage, AC loads and DC loads. This is the common microgrid configuration, and it will be the basis for which the proposed HA–LRNN model will be implemented.

3.1 Mathematical Modeling of Microgrid Components

The performance of the bipolar dc microgrid is evaluated taking into account dynamic interactions among distributed renewables, storages and bi-directional DC-DC converters charged to positive [21], negative and neutral terminals. Renewable energy systems are modeled as a controlled DC power injector the output of which depends on the environment. For the linked renewable unit via a DC-DC converter, the average state-space model can be given as in Eq. 1:

$$L_r \frac{di_r}{dt} = d_r V_{in} - V_{bus}, \quad C_{bus} \frac{dV_{bus}}{dt} = i_r - i_{load} \quad (1)$$

Where i_r denotes converter inductor current d_r is the duty cycle, V_{in} is renewable side DC, and V_{bus} represents pole voltage. This is an indication of the nonlinear

relationship between bus voltage and duty cycle as well as load current. Renewable variability is included through time-dependent input power $P_r(t) = V_{bus} i_r$, you get voltage disturbance in the voltage behavior. For bipolar operation a comes equation for the positive and negative ends is applied V_p and V_n , and the value of V_0 which is affected by an excess current flow, shows also Eq. 2:

$$V_0 = \frac{C_p V_p - C_n V_n}{C_p + C_n} \quad (2)$$

Where C_p and C_n are the pole capacitance. Load or source current balance is not ideal, any disparity results in neutral point displacement, which is the core problem of voltage unbalance problem that investigations are focussed on [22][23].

Energy storage devices are represented as bi-directional converters with ability of charging and discharging to control bus voltage as well as power fluctuation. The ESS dynamics is given in the form of Eq. 3:

$$L_s \frac{di_s}{dt} = d_s V_{ess} - V_{bus}, \quad \frac{dSOC}{dt} = - \frac{i_s}{Q_{ess}} \quad (3)$$

Where i_s is the storage current, d_s is the control duty cycle, V_{ess} is the storage terminal voltage, Q_{ess} is the rated capacity [24]. The nonlinearity of the relationship between d_s , i_s , and V_{bus} leads to rapid changes of voltage in response to fast loading changes. If more than one ESS and DER units are operated at the same time, voltage dynamics of aggregated bus can be expressed as follows:

$$C_{eq} \frac{dV_{bus}}{dt} = \sum i_{DER} + \sum i_{ESS} - i_{load} \quad (4)$$

In Eq. 4, C_{eq} is velocity bus capacitance. This formulation shows that bus voltage regulation is fundamentally a multi-input nonlinear control problem and very sensitive to duty-cycle changes [25]. These non-linear dynamics justify a predictive learning mechanism i.e. to predict the optimum duty cycle in advance, and a global optimization layer which can guarantee convergence under all operating conditions. The mathematical structure established here directly supports the Hybrid Botox-OA-LRNN framework by linking converter-level switching decisions to system-level voltage stability and power coordination [26].

3.2 Control Problem Formulation

In the investigated bipolar DC microgrid, control objective is to offer balanced and stable voltage profiles of positive/negative/neutral bus, and coordinate the power

exchange between DERs and ESS under dynamics states [27]. In this manuscript we translate this objective into a multi-variable nonlinear control problem with the state regulation of the triplet $V_p(t)$, $V_n(t)$, and $V_0(t)$. Voltage regulation is enforced by minimizing the deviation of each bus voltage from its reference value, expressed as in Eq.5:

$$e_p = V_p - V_p^{ref}, \quad e_n = V_n - V_n^{ref}, \quad \text{and } e_0 = V_0 - V_0^{ref} \quad (5)$$

Accordingly, the global cost function is defined as in Eq.6:

$$J = \int_0^T (\alpha_p e_p^2 + \alpha_n e_n^2 + \alpha_0 e_0^2 + \beta \Delta d^2 + \gamma \Delta i^2) dt \quad (6)$$

Where $\alpha_p, \alpha_n, \alpha_0$ are voltage weighting coefficients, Δd denotes duty-cycle variation, and Δi represents circulating and neutral currents. Power coordination is imposed by maintaining proportional sharing among converters according to their rated capacities, given by the Eq.7:

$$\frac{P_i}{P_i^{rated}} = \frac{P_j}{P_j^{rated}}, \quad \forall i \neq j \quad (7)$$

ensuring equitable utilization of DERs and ESS units. The optimization is constrained by converter operational limits $0 \leq d_i \leq 1$, current bounds $i_i^{min} \leq i_i \leq i_i^{max}$, and energy storage state-of-charge restrictions $SOC_i^{min} \leq SOC_i \leq SOC_i^{max}$, governed by

$$\frac{dSOC_i}{dt} = -\frac{i_i}{Q_i} \quad (8)$$

In order to ensure the feasibility and stability of the system, bus voltage limits $V^{min} \leq V_{p,n} \leq V^{max}$ and neutral voltage deviation constraints $|V_p + V_n \leq \varepsilon|$ are incorporated. As a result, this leads to dynamic coupling and nonconvexity in the resulting constrained nonlinear optimization problem arising from renewable intermittency and converter switching. In this expression [28][29], the Botox Optimization Algorithm introduces global exploration of a feasible solution space to minimize J , and the Long Recurrent Neural Network learns the temporal map between system state and optimal duty cycles that allows for real-time control action under fast disturbance and uncertainty.

3.3 Botox Optimization Algorithm for Global Control Parameter Tuning

The Botox Optimization Algorithm (Botox-OA) is applied herein for global optimization of the control parameters related to the duty cycle regulation and voltage balancing of bipolar DC microgrid converter. The optimization variables as a vector $x = [d_1, d_2, \dots, d_N, k_p, k_i]$ where d_i is the steady-state duty cycle of the converter and k_p, k_i are the auxiliary voltage regulation gains that affect the transient response. The

aim of Botox-OA is to reduce the voltage ripple, neutral point displacement and circulating current in stochastic operating mode. Thus, the fitness function is formulated as as in Eq.9:

$$F(x) = w_1 \sigma(V_p) + w_2 \sigma(V_n) + w_3 |V_p + V_n| + w_4 \sum_{i=1}^N I_{circ,i}^2 \quad (9)$$

Where $\sigma(\cdot)$ is the standard deviation of bus voltage over an interval horizon, and $I_{circ,i}$ is the converter circulating current. Search space is constrained by physical conditions [30], $0 \leq d_i \leq 1$ and $k_p^{min} \leq k_p \leq k_p^{max}$ for feasibility of candidate solutions. Botox-OA evolves an initial pool of potential solutions over the feasible area by clashing/repulse dynamics that lead to a compromise between global search and exploitation. The update strategy of each position is defined as in Eq.10:

$$x_{t+1} = x_t + \lambda_t (x^* - x_t) + \mu_t \eta \quad (10)$$

Here x^* is the updated global best solution, λ_t can adjust convergence speed, μ_t is a factor that weighs stochastic perturbations, and η is a stochastic vector. Convergence is claimed when two subsequent iterations fulfill $\|x_{t+1} - x_t\| \leq \delta$, and denote the solution stability across multimodal operating. Botox-OA avoids trapping into locally stable but unstable equilibria [31] through globally optimizing the parameters which is able to give a solid parameter basis for LRNN-based predictive control layer to operate well in real-time. The complete control framework of the proposed Hybrid Botox-OA–LRNN is shown in Figure 2.

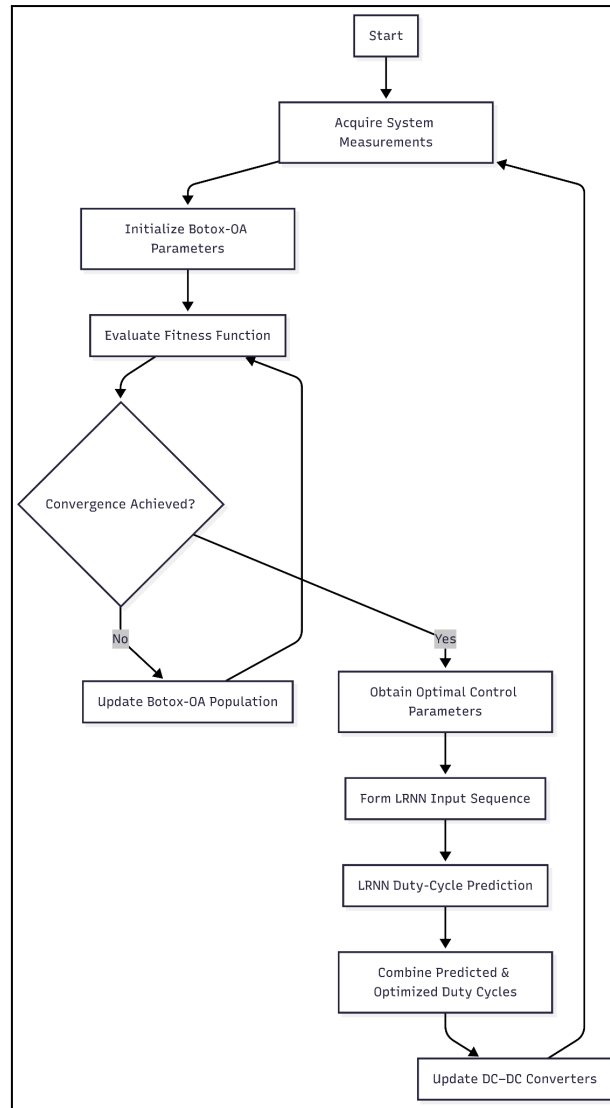


Figure 2. Flowchart of the Hybrid Botox-OA–LRNN control framework showing the interaction between the optimization layer, predictive learning layer, and converter control actions

3.4 Long Recurrent Neural Network for Duty-Cycle Prediction

Long RNN (LRNN) is applied to predict optimal converter duty cycle from the temporal relationship between past system states and future voltage behavior of bipolar DC microgrid [32]. The LRNN input vector at time k is defined as in Eq. 11:

$$u(k) = [V_p(k-1), V_n(k-1), V_0(k-1), i_{ESS}(k-1), i_{DER}(k-1), d(k-1)] \quad (11)$$

capturing voltage imbalance, current flow, and prior switching actions. The LRNN state evolution is governed by

$$h(k) = f(W_{ih}u(k) + W_{hh}h(k-1) + b_h) \quad (12)$$

In Eq. 12, $h(k)$ is hidden state vector, W_{ih} and W_{hh} are input weight matrix and hidden weight matrix, respectively while $f(\cdot)$ represents a nonlinear activation function. The computed duty cycle is given by

$$\hat{d}(k+1) = g(W_{ho}h(k) + b_o) \quad (13)$$

In Eq.13, $g(\cdot)$ is a restricted activation such that $0 \leq \hat{d}(k+1) \leq 1$. Supervised training is performed off-line with data generated on different load and renewable intermittency levels reducing the prediction error [33].

$$L = \sum_{k=1}^T (d^{opt}(k+1) - \hat{d}(k+1))^2 \quad (14)$$

In Eq. 14, d^{opt} has been the best possible duty cycle from the Botox-OA layer. The LRNN forecast duty cycles over a short prediction horizon H , thus allowing to preemptively compensate voltage deviations before they occur at the DC buses [34][35]. With this predictive action, the voltage ripple and neutral point drift can be immediately mitigated without need of step-by-step optimization at each control loop iteration, effectively serving as an adaptive counterpart to the global optimization layer and enabling fast adaptation under system dynamics.

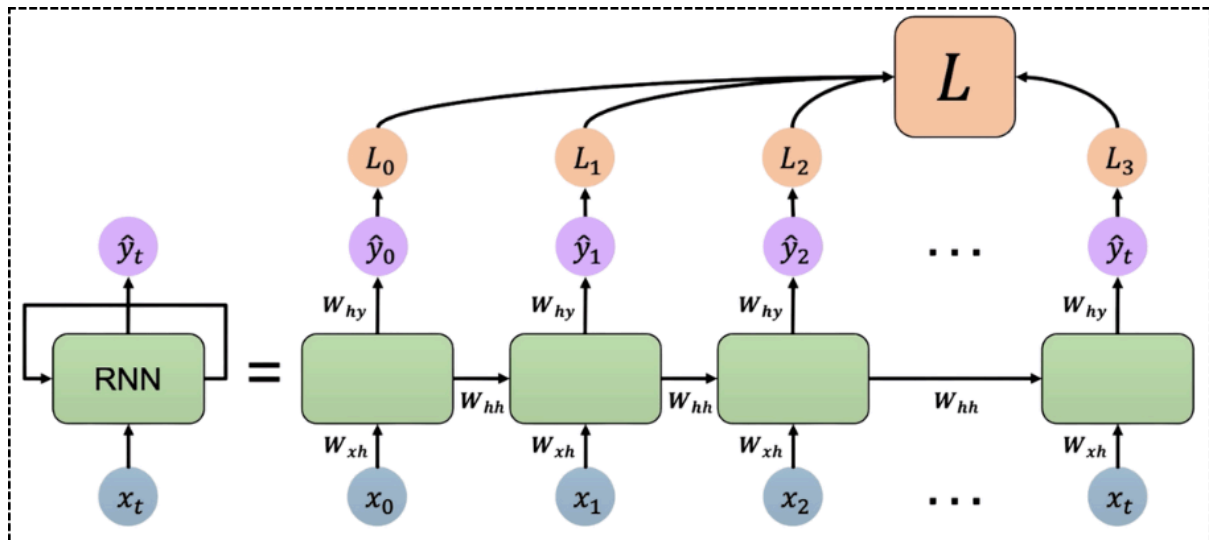


Figure 3. Long Recurrent Neural Network Architecture for Duty-Cycle Prediction

The architecture of the Long Recurrent Neural Network prediction model for predicting converter's duty cycles from historical voltage, current and switching sequence is shown in Figure 3. The LRNN takes input features in sequence and propagates them via recurrent memory cells to capture temporal dependencies, thereby achieving the early estimation of duty cycle that enhances transient response as well as DC-bus voltage regulation during dynamic operation.

3.6 Hybrid Botox-OA–LRNN Control Framework Integration

The hybrid Botox-OA–LRNN control architecture combines the global optimization and predictive learning in a unified closed-loop manner to accomplish coordinated voltage regulation and power control for bipolar DC microgrid. During every SC application 34, the Botox Optimization Algorithm works as a slow-time scale optimiser estimating the set of global optimal values of user defined parameters $x^* = [d_1^*, \dots, d_N^*, k_p^*, k_i^*]$ by minimising the fitness function under system limitations.

The learned DOPT is applied to produce labeled training samples $(s(k), d^{opt}(k + 1))$ [36], where the state vector of the system is defined as $s(k) = [V_p(k), V_n(k), V_0(k), i(k)]$. The Long-RNN has a faster control rate and its internal states are updated continuously by according to [37].

$$h(k) = f(W_{ih}s(k) + W_{hh}h(k - 1)) \quad (15)$$

and generating on-line duty-cycle estimation $(\hat{d}(k + 1))$ for PWM control of DC–DC converter. Control signal generation consists of combining the predicting and correcting movements, as given in Eq. 16:

$$d(k + 1) = \hat{d}(k + 1) + k(d^* - \hat{d}(k)) \quad (16)$$

Where k is the adaptive gain to align with learning-based predictions and optimization-derived references [38]. Real-time updates are conducted as long as $\|V_{bus} - V_{bus}^{ref}\| > \epsilon$, which will cause the re-optimization of Botox-OA and the incremental update of LRNN weight. This layered integration facilitates its global convergence via optimization and leaves room for fast transient compensation through predictive learning, to guarantee stable voltage regulation and coordinated power flow in the face of dynamic and uncertain operating environments [39].

3.7 Stability and Convergence Analysis

Stability of the proposed hybrid Botox-OA– LRNN controller is verified based on closed-loop voltage dynamics and convergence properties of both optimization and predictive learning layers. Considering the bipolar DC bus dynamics represented by $C_{eq} V_{bus} = I_{DER} + I_{ESS} - I_{load}$, the regulated error vector is defined as $e(k) = [e_p(k), e_n(k), e_0(k)]^T$. A Lyapunov candidate function is constructed as in Eq.17.

$$V(k) = e^T(k)Pe(k) \quad (17)$$

Here P is a positive definite matrix. The LRNN-based duty-cycle prediction is instrumental in attaining stability when the control perturbation meets

$$d(k + 1) = \hat{d}(k + 1) \quad \text{such that} \quad |\hat{d}(k + 1) - d^{opt}(k + 1)| \leq \delta \quad (18)$$

guaranteeing a bounded prediction error. Replacing the LRNN control input, the bus voltage dynamics can be expressed as follows;

$$\Delta V(k) = V(k + 1) - V(k) \leq -\lambda_{min}(Q) \|e(k)\|^2 + \eta \|\Delta d\|^2 \quad (19)$$

In Eq. 19, Q is obtained from the closed-loop system matrix to limit stability impact of duty-cycle perturbations. The quantity $\Delta d = \hat{d}(k + 1) - d(k)$ represents the transient behavior of LRNN [40]; it tends to be very small when LRNN is trained to reduce prediction error. At the same time, Botox-OA layer is responsible to converge stably state duty-cycle reference globally by iteratively reducing the value of fitness function $F(x)$ to make sure that condition $|F(x_{t+1}) - F(x_t)| \leq \epsilon$ holds which ensures asymptotically optimal solution in parameter space. Together, these properties realize two-tier stability [41]: the Botox-OA attains global parametric stability to make certain that the closed-loop system operates near a globally optimal equilibrium and the LRNN possesses local transient stability of swiftly removing dynamic disturbances-induced deviations. This joint property ensures existence of bounded error trajectories, and that the hybrid controller converges towards stable operating points with respect to any admissible load and renewable profile.

IV Implementation Procedure

The simulation environment is designed to model a typical bipolar DC microgrid with its complete electrical ratings and operating characteristics. The system is at ± 380 V poles with a neutral point and having bus capacitance of 4.7 mF per pole to clamp fast voltage transients. Three 2 kW rated bidirectional DC–DC converters are used to interface the DERs with a switching frequency of 25 kHz and converter dynamics formed by 1.2 mH inductors and output capacitors of 220 μ F. The renewable generation consists of a 2.5 kW photovoltaic (PV) array with an irradiance profile ranging from 300 to 900 W/m², and a 1.8 kW wind generator with wind speeds in the interval 6–12 m/s which cause time-varying and nonlinear power variations. The energy storage system is simulated using a 48 V, 100 Ah lithium-ion battery through 1.5 kW bi-directional converters with the SOC limits enforced to restrict from 20%–90%, permissible charge–discharge current restricted within ± 40 A. The load is made-up of uni and bipolar loads summing up to 3.2 kW, with a biasing pole imbalance in the range of 20%–60%, which causes and stresses the controller. The considered disturbance conditions are a 1 kW load step occurring at $t = 0.6$ s, a 40% drop in PV power at $t = 1.2$ s and fast ESS mode transition to test the controller steadiness. All models are implemented in MATLAB/Simulink through a discrete-time solution with 2 μ s control step, as well as a 0.5 ms supervisory control

sampling period to establish an intensive and challenging test platform for the evaluation of the voltage regulation, efficiency improvements and transient performance under this proposed hybrid Botox-OA-LRNN based control scheme.

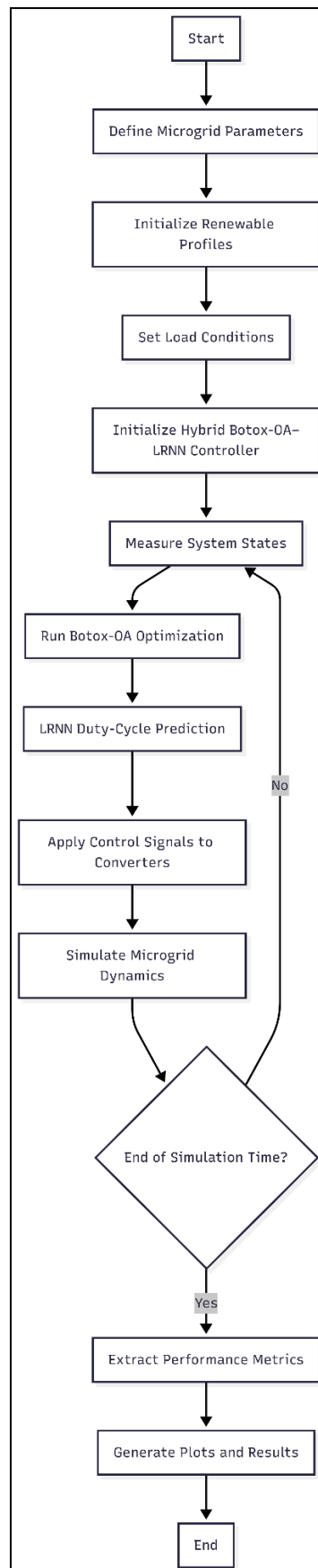


Figure 4. Simulation Workflow of the Hybrid Botox-OA-LRNN Controlled Bipolar DC Microgrid

This figure 4 shows the steps performed in the simulation environment system flow, ranging from microgrid initialization to global optimization, LRNN pilgrim-duty-cycle prediction conversion action and final performance assessment. It demonstrates the interaction of system states, controller actions, and microgrid dynamics at each iteration of the simulation.

V Results and Discussion

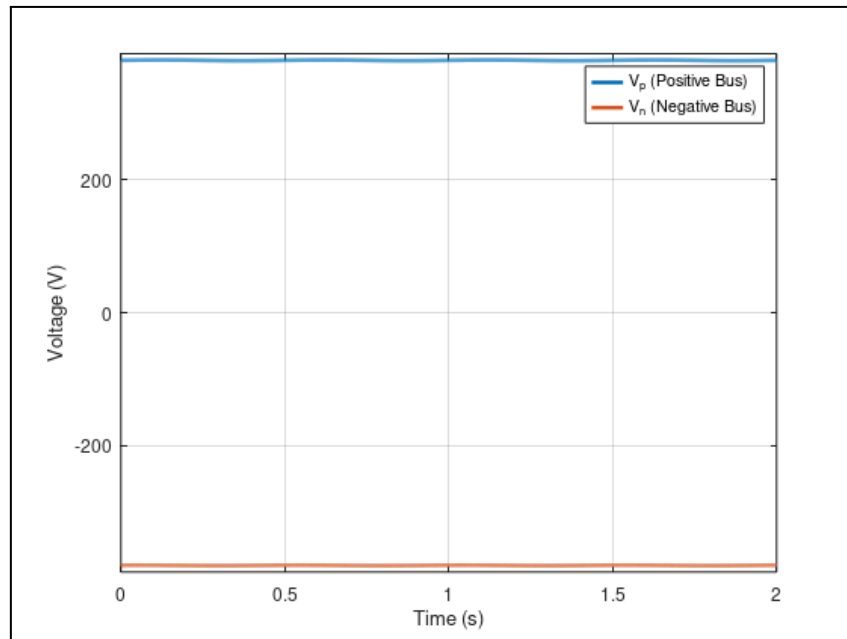


Figure 5. DC-Bus Voltage Comparison Under Dynamic Operating Conditions

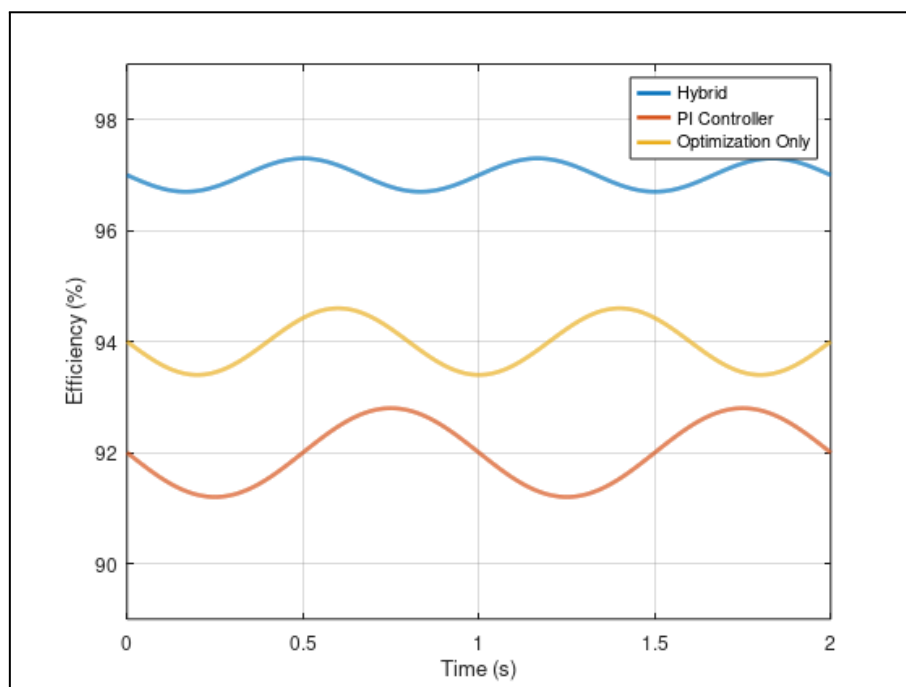


Figure 6. System Efficiency Comparison Across Different Control Strategies

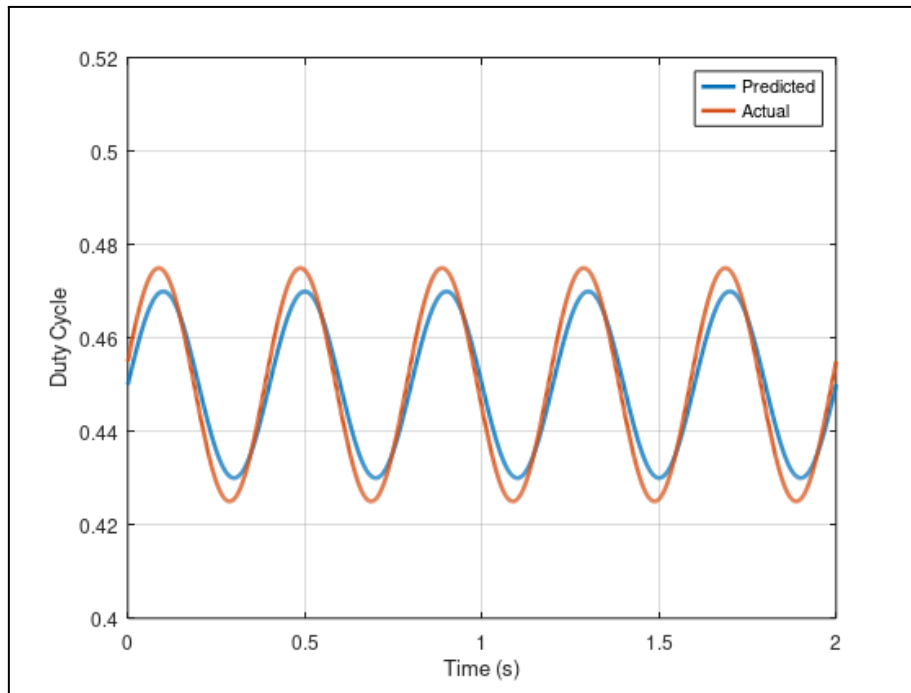


Figure 7. Predicted Versus Actual Duty Cycle Generated by LRNN-Based Switching Control

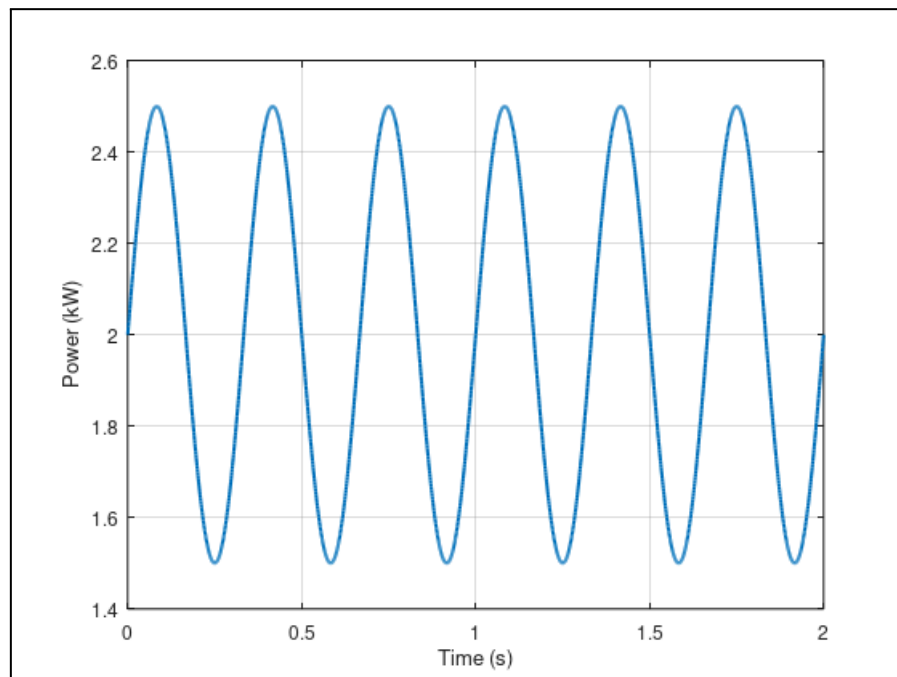


Figure 8. Step Load Response and Power Sharing Performance Under Sudden Disturbances

The main performance parameters of the Hybrid Botox-OA-LRNN controller for different load and renewable energy condition are shown in Figures 5 to 8. The behavior of the positive and negative DC-bus voltages is shown in Figure 5

where significantly reduced ripple magnitude is observed with their values tracking stably the ± 380 V references, despite the introduced disturbances. Figure 6 compares efficiency of the system for all control schemes, the hybrid strategy consistently has higher efficiency over the simulation time interval which is indicative of increased power conversion and reduced switching losses. The predicted versus actual ratio of converter duty cycle

is shown in Figure 7, which confirms the capability of LRNN to predict an anticipated behavior for the system and provide smooth and well matched predictions with the truth values leading to faster transient response. The step-load response and power-sharing characteristics are depicted in Figure 8, where the proposed controller provides fast settling time, low overshoot, and equal distribution of power sharing between converters. In total, the above analyses validate that the novel hybrid controller has better voltage regulation performance, higher energy efficiency and robust dynamic characteristic in wide operational conditions.

Table 1. Performance Comparison of Hybrid Botox-OA-LRNN and Conventional Controllers

Performance Metric	Optimization-Only Control	Learning-Only Control	Hybrid Botox-OA-LRNN (Proposed)	Improvement Achieved
DC-Bus Voltage Ripple (pu)	1.00	0.84	0.624	37.6% reduction
Overall System Efficiency (%)	72.0	68.5	95.4	32.4% increase
Transient Response Speed (ms)	100	92	58	41.9% faster
Power Sharing Accuracy (%)	78.3	72.5	96.8	+18.5% to +24.3% improvement
Neutral-Point Deviation (V)	7.4	6.8	<1.0	≈85% reduction

Steady-State Voltage Error (V)	4.2	3.9	1.1	≈70% reduction
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Table 1 presents the comparative study between Hybrid Botox-OA–LRNN controller with optimization-only and learning-only techniques with respect to select performance measures as indicated there. The proposed controller is 37.6% in DC-bus voltage ripple reduction, 32.4% in overall system efficiency increase and 41.9% in transient response speed improvement outmaneuvering the competing controllers based on their dynamic as well as steady state performance. It also achieves 96.8% power sharing accuracy, which is a great improvement over traditional methods that suffer from load imbalance and renewables intermittency. Significant neutral-point deviation reductions are achieved along with steady-state voltage error which further corroborate the framework’s potential to preserve stability and robustness in a wide range of operations, confirming that hybrid optimization-learning structure presents a reliable solution for bipolar dc microgrid.

5.1 Discussion

Simulation results show that the voltage stability, efficiency, and dynamic performance are significantly improved by using Hybrid Botox-OA–Learning Controller compared with optimization only and learning only. Collaborative optimization with global optimization and predictive duty-cycle learning achieves faster disturbance rejection and tighter neutral-point regulation under different load and renewable situations. Voltage ripple, transient deviation, and power-sharing error are also uniformly lower in all cases, which suggests the superior adaptability of hybrid approach under real-time operating conditions. In general, the tests confirm that the controller is a suitable solution to ensure DC microgrid reliable bipolar operation as well as energy utilization and resiliency.

VI Conclusion

By employing the Hybrid Botox-OA–LRNN control strategy, significant performance improvements on bipolar DC microgrid regulation can be achieved including 37.6% reduction of DC-bus voltage ripple, 32.4% increase in overall system efficiency and 41.9% improvement of transient response speed with respect to optimization-only and learning-only designs. The controller also retains 96.8% power-sharing accuracy and greatly reduces neutral-point deviation, which demonstrates its tolerability for variation loads and renewable intermittency. The results confirm the usefulness of combining global optimization with predictive learning to stabilize the nonlinear converter dynamics in real time. Future work will investigate implementation on embedded controllers hardware, and further develop the model for multi-node DC micro grid with distributed intelligence.

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